

GNN–BERT for Multimodal Music Understanding and Cross-Modal Retrieval

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Abstract—Understanding music requires both acoustic information from audio and semantic information expressed through language. However audio-only and text-only approaches capture only part of the musical context. This study investigates a multimodal framework which combines graph-based audio representations with pretrained language representations for music understanding and cross-modal retrieval. Audio recordings are represented as temporal graphs and processed using GraphSAGE while DistilBERT encodes natural-language music descriptions from MusicCaps. The study evaluates individual modalities and multimodal fusion by using early concatenation and cross-attention. The cross-attention model achieves the strongest supervised performance with 80.52% accuracy, 73.79% Macro-F1 and 91.88% Macro AUC-PR. A separate contrastive graph–text model using symmetric InfoNCE loss learns a shared embedding space for cross-modal retrieval and zero-shot genre prediction remains challenging. Overall the results demonstrate the potential of combining structured audio representations with textual semantic information for multimodal music understanding and retrieval.

Index Terms—Multimodal music understanding, Graph neural networks, GraphSAGE, DistilBERT, MusicCaps, Cross-modal retrieval, Contrastive learning

I. INTRODUCTION

A. Background and Motivation

Music understanding is inherently multimodal. A musical piece can be characterized not only by its acoustic properties such as rhythm, timbre, harmony, instrumentation and temporal structure but also by textual descriptions that capture higher-level semantic information such as genre, mood, instruments, performance characteristics and listening context. Traditional music classification systems often focus on audio signals mainly while language-based approaches rely on captions, tags, lyrics or other textual descriptions. Although these approaches can be effective individually, each modality provides only a partial representation of the underlying musical content.

Recent advances in deep learning have enabled more expressive representations of both music and language. In particular, graph neural networks (GNNs) provide a natural framework for representing structured musical information. Audio can be segmented into temporally related units and represented as graphs, where nodes represent musical segments and edges represent relationships between neighboring segments. This allows models such as GraphSAGE to learn representations

that incorporate relationships between neighboring musical segments. At the same time, pretrained language models such as BERT and DistilBERT can encode textual descriptions into semantic representations that capture relationships between words, phrases and broader musical concepts.

Combining these modalities provides an opportunity to obtain a more comprehensive representation of music. A graph-based audio representation can provide information about acoustic and structural characteristics and a language representation can provide semantic context. This motivates the development of a GNN–BERT framework that investigates whether the complementary information contained in audio structure and textual descriptions can improve music understanding and enable cross-modal retrieval.

This study is structured around four main tasks to investigate this multimodal framework systematically. First, a textual baseline is established using natural-language music descriptions to evaluate text-based music understanding. Second, a graph-based audio representation is evaluated against a conventional spectrogram-based baseline for genre classification. Third, multimodal fusion mechanisms are explored to integrate graph-derived audio representations with textual representations. Finally, contrastive learning is applied to map graph and textual representations into a shared embedding space for cross-modal retrieval.

B. Problem Statement

Audio-only and text-only music understanding approaches have important limitations when used independently. Audio-based models can capture acoustic characteristics and musical structure but they have difficulty representing high-level semantic concepts that are naturally expressed through language. Text-based models can capture semantic information from captions and tags but they do not directly observe the acoustic or structural characteristics of a musical recording.

A further challenge is that musical structure is not always properly represented by treating an audio signal as a simple sequence. Relationships between neighboring musical segments, changes in acoustic characteristics and local structural dependencies can be represented more explicitly using graph-based representations. However, a graph-only representation may still lack the semantic information available from textual descriptions.

Another challenge for the multimodal experiments is the lack of direct instance-level correspondence between the GTZAN and MusicCaps datasets. The multimodal experiments use genre-based cross-dataset pairing rather than claiming that a particular GTZAN recording corresponds directly to a particular MusicCaps caption. This allows the study to examine category-level multimodal learning and also avoid unsupported claims of instance-level correspondence.

So this research investigates how graph-based representations of musical audio can be combined with pretrained language representations of textual music descriptions to improve music understanding and enable cross-modal representation learning and retrieval.

C. Research Objectives

The primary objective of this study is to investigate multimodal approaches for music context understanding by combining structured audio representations with textual music information. The specific objectives are as follows:

- 1) Develop a BERT-based text baseline for music tag and context understanding using natural-language music descriptions.
- 2) Construct audio-derived segment graphs and evaluate a GraphSAGE model against a conventional CNN mel-spectrogram baseline for music genre classification.
- 3) Develop a GNN–BERT multimodal fusion framework to integrate graph-based audio features with textual representations and compare different fusion strategies.
- 4) Develop a contrastive GNN–BERT model for learning a shared representation space between graph-based audio and textual representations for cross-modal retrieval.
- 5) Analyze model performance and learned representations using quantitative and qualitative evaluation.

II. RELATED WORKS

A. Music Representation Learning

Music representation learning aims to transform raw or processed audio signals into representations that can be used for tasks such as music tagging, classification and retrieval. Spectrogram-based convolutional neural networks have been widely used because they can learn local time-frequency patterns from music signals.

Choi et al. [1] investigated automatic music tagging using fully convolutional neural networks with mel-spectrogram inputs. Their experiments on MagnaTagATune and the Million Song Dataset demonstrated the effectiveness of convolutional models for learning music representations from time-frequency features.

Choi et al. [2] later introduced convolutional recurrent neural networks (CRNNs) for music tagging, combining convolutional layers for local feature extraction with recurrent layers for temporal summarization. Their results demonstrated the value of modeling both local acoustic patterns and longer-term temporal relationships in music.

More recently, Spijkervet and Burgoyne [3] proposed Contrastive Learning of Musical Representations (CLMR), a self-supervised framework that learns musical representations directly from raw waveforms without requiring manually annotated labels during representation learning. The learned representations demonstrated useful performance on downstream music classification and tagging tasks.

These studies show the progression from conventional spectrogram-based feature learning toward models that capture temporal structure and learn more general representations from unlabeled audio. However, these approaches primarily represent music through time-frequency patterns, feature sequences or learned embedding spaces rather than modeling relationships between individual audio segments as graphs. This motivates alternative representations that model structural relationships across audio segments explicitly.

B. Graph Neural Networks for Music

Graph neural networks (GNNs) are designed to learn from data represented as graphs where information is defined by both node features and relationships between nodes. Kipf and Welling [4] proposed Graph Convolutional Networks (GCNs), which learn node representations by aggregating information from neighboring nodes while incorporating local graph structure. Although not designed specifically for music, GCNs provide a foundation for learning representations from structured relationships.

Hamilton et al. [5] introduced GraphSAGE which is an inductive graph representation learning framework that generates node embeddings by sampling and aggregating information from local neighborhoods. Its ability to learn representations from node features and neighborhood structure makes it suitable for graph-based learning with varying graph structures.

GNNs have also been applied directly to musical data. Jeong et al. [6] applied graph neural networks to expressive piano performance by representing music scores as note-level graphs. Their results showed that the graph-based representation could improve expressive piano performance compared with models that treated the music score as a one-dimensional sequence.

These studies demonstrate the ability of GNNs to model relationships within structured data and musical representations. However, graph-based audio representations alone may not capture the semantic information available in textual music descriptions which motivates the integration of GNN-based audio representations with language models.

C. Transformer-Based Music and Text Understanding

Transformer-based models have become widely used for learning contextual representations from text. Vaswani et al. [7] introduced the Transformer architecture which uses attention mechanisms to model relationships between tokens without relying on recurrent or convolutional layers. The Transformer provided the foundation for later pretrained language models such as BERT.

Devlin et al. [8] introduced BERT which is a pretrained language representation model that learns bidirectional contextual representations by considering both left and right context. Its contextual representations make BERT suitable as a general text encoder for natural-language music descriptions.

Sanh et al. [9] proposed DistilBERT which is a smaller and more computationally efficient version of BERT trained using knowledge distillation. It retains the ability to produce contextual text representations while reducing model size and inference cost which makes it suitable for the text-based models used in this study.

These works establish the progression from the Transformer architecture to pretrained contextual language models and efficient variants such as DistilBERT. Such models provide the textual representation which is needed to capture semantic information from natural-language music descriptions and complement the graph-based audio representations discussed in the previous section.

D. Multimodal and Contrastive Learning

Multimodal learning aims to combine information from different modalities such as audio and text, to learn representations that contain complementary information. In music-related applications, joint audio-text representations can support tasks such as music tagging, semantic understanding and cross-modal retrieval.

Huang et al. [10] proposed MuLan which is a joint embedding model that connects music audio with natural-language descriptions. The model uses separate audio and text encoders and learns to represent related audio and text in a shared embedding space. Their results demonstrate the usefulness of natural-language descriptions as semantic supervision for music understanding and cross-modal retrieval.

Elizalde et al. [11] introduced CLAP (Contrastive Language-Audio Pretraining) which learns joint audio-text representations using contrastive learning. The model brings matching audio-text pairs closer in the embedding space while separating non-matching pairs. CLAP showed the potential of contrastive learning for zero-shot classification and audio-text retrieval across different audio domains including music.

Agostinelli et al. [12] introduced MusicLM which is a text-conditioned music generation system that uses natural-language descriptions to control music generation. The work also introduced MusicCaps which is a dataset of 5.5k music-text pairs with rich natural-language descriptions which are provided by human experts which provides the textual resource used in this study.

These studies demonstrate the potential of combining audio and language representations for music understanding. MuLan and CLAP especially motivate the use of shared audio-text embedding spaces and contrastive learning for cross-modal retrieval. Building on these ideas this study combines a graph-based audio representation with a pretrained language model and investigates both multimodal fusion and contrastive cross-modal learning.

E. Research Gap

Existing studies have demonstrated effective music representations using convolutional and contrastive audio models, graph neural networks for structured musical data and transformer-based models for textual representations. Multimodal approaches such as MuLan and CLAP have further shown the potential of aligning audio and natural-language representations. However within the literature reviewed here, the combination of graph-based audio representations with transformer-based textual representations for multimodal fusion and cross-modal retrieval has received comparatively less attention in existing music understanding research.

So this study investigates whether graph-based audio representations and transformer-based textual representations can be used jointly within a unified music-understanding framework. The study addresses this gap through GNN-BERT multimodal fusion and contrastive learning for cross-modal retrieval.

III. METHODOLOGY

A. Overall Framework

This study develops a multimodal framework that combines audio-based graph representations with textual representations to investigate music context understanding. The overall framework uses two main data sources: GTZAN for audio and MusicCaps for natural-language music descriptions.

The audio branch starts with the GTZAN recordings. The audio files are preprocessed and divided into suitable segments, after which audio features are extracted from the segments. These features are then used to construct graphs that represent relationships between different parts of the music. A GraphSAGE-based graph neural network is used to learn representations from these graphs. The text branch uses the descriptions from MusicCaps. The text is tokenized and processed using a pretrained DistilBERT-based language model. The resulting contextual embeddings provide semantic representations of the music descriptions.

GTZAN and MusicCaps do not provide direct instance-level correspondence between individual audio recordings and textual descriptions so the multimodal experiments use a genre-based pairing strategy. Graph representations from GTZAN and textual representations from MusicCaps are paired according to their genre rather than treating a specific MusicCaps description as the description of a specific GTZAN recording. This pairing strategy allows the study to investigate multimodal learning while avoiding unsupported claims of direct correspondence between individual audio and text samples.

The framework is evaluated through four main tasks. First, a DistilBERT-based text-only model is developed as a baseline for music context understanding. Second, the graph-based audio representation is evaluated using a GraphSAGE model for genre classification and compared with a CNN-based mel-spectrogram baseline. Third, the graph and text representations are combined using multimodal fusion methods including early concatenation and cross-attention. Finally, contrastive learning with InfoNCE loss is used to learn a shared representation space between graph-based audio representations

and textual descriptions, which is then evaluated through cross-modal retrieval.

The complete workflow consists of data preparation, audio feature extraction, graph construction, text processing, unimodal model training, genre-based multimodal pairing, multimodal fusion and contrastive representation learning. The following sections describe each stage of the framework in detail.

B. Datasets

This study uses two main datasets, GTZAN and MusicCaps. They provide audio and textual information for different parts of the proposed framework.

GTZAN is an audio dataset containing 1,000 music tracks from 10 different genres: blues, classical, country, disco, hip-hop, jazz, metal, pop, reggae and rock. In this study, GTZAN is used as the primary audio dataset. The recordings are processed to extract audio features, which are then used to construct music graphs for the graph-based models. The genre labels are used for the genre classification experiments and for evaluating the GraphSAGE model against the CNN-based spectrogram baseline.

MusicCaps is a dataset of approximately 5,500 music clips paired with natural-language descriptions. The descriptions provide natural-language information about musical characteristics such as instruments, tempo, mood, genre and other aspects of the music. In this study, MusicCaps is used as the primary textual dataset. The descriptions are processed using a DistilBERT-based language model to obtain contextual text representations for the text baseline and multimodal experiments.

C. Audio Preprocessing

The GTZAN audio recordings were preprocessed before feature extraction and graph construction. All recordings were loaded as mono audio and resampled to 22,050 Hz to maintain a consistent sampling rate across the dataset.

Each track was then normalized using peak normalization. The audio signal was divided by its maximum absolute amplitude to normalize the peak level before feature extraction. After normalization, each track was divided into fixed-length 5-second segments. Segments shorter than 5 seconds at the end of a track were discarded. This segmentation allowed different parts of a track to be represented separately and later used as individual nodes in the music graph. For each segment, two types of audio features were extracted: a 128-bin log-mel spectrogram and 12-bin chroma features. The log-mel spectrogram captures the time-frequency characteristics of the audio and the chroma features represent the distribution of energy across the 12 pitch classes.

The mean was calculated across the time dimension of each feature matrix in order to obtain a fixed-length representation for each segment. The resulting mel and chroma vectors were then concatenated to form a single feature vector for each audio segment. These segment-level feature vectors were used as the node features during graph construction.

D. Music Graph Construction

Each GTZAN recording was represented as a temporal graph where the nodes correspond to consecutive 5-second segments of the recording. The 140-dimensional feature vector of each segment, that is obtained by combining its pooled mel-spectrogram and chroma features, was assigned to the corresponding node.

The nodes were connected according to their temporal order in the original recording. Each node was connected to the immediately following node, creating edges between consecutive segments. This produces a simple temporal graph that represents how the acoustic characteristics change throughout a music track.

The resulting graphs were stored as NetworkX graph objects. For GraphSAGE training, the graphs were later converted into PyTorch Geometric graph objects. The node feature vectors were used as the node feature matrix and the graph-level genre was converted into a numerical label representing one of the ten GTZAN genres. This graph representation provides the structured audio input used by the GraphSAGE model for genre classification.

E. Text Processing and Representation

MusicCaps captions were used as the textual input to the framework. The captions were tokenized using a tokenizer corresponding to the pretrained DistilBERT model. Padding and truncation were applied with the maximum input length set to 256 tokens during model training. A pretrained DistilBERT language model was used to encode the captions into contextual text representations. The representation of the first token was used as the text embedding. These embeddings provide a semantic representation of the information that is contained in each music description.

F. DistilBERT Baseline for Music Tag Understanding

A text-based baseline was developed using the natural-language descriptions from the MusicCaps dataset. The purpose of this baseline was to evaluate how well textual music descriptions can be used to predict musical characteristics without using the graph-based audio representation.

Each caption was represented using a multi-label target consisting of the 50 most frequent musical tags in the dataset. A binary label vector was created for each caption, where each position indicates whether the corresponding tag was associated with the caption. So a single caption could have multiple positive labels. The tokenized captions were provided to the pretrained DistilBERT model to obtain contextual representations of the text. The representation of the first token was used as the text embedding and was passed through a linear classification layer to produce predictions for the 50 tags. Binary cross-entropy with logits loss was used to train the multi-label classifier. The model was fine-tuned using the AdamW optimizer for five epochs. The validation set was used to monitor performance during training, and the model with the highest validation Macro-F1 score was saved as the best model.

After training, the best model was evaluated on the test set using Macro-F1, Micro-F1, Macro Precision, Macro Recall and test loss. These measures were used to evaluate the model’s performance across the selected set of musical tags.

G. GraphSAGE and CNN Baseline

The graph-based audio representation was evaluated using a GraphSAGE model for music genre classification. Each input graph contained 140-dimensional node features, with each node representing a 5-second segment of a GTZAN recording. The GraphSAGE model used two SAGEConv layers to learn node representations by aggregating information from neighboring segments in the graph. ReLU activation was applied after each graph convolution, followed by dropout to reduce overfitting. After the graph convolution layers, global mean pooling was used to combine the node representations into a single representation for each music track. This graph-level representation was then passed through a linear classification layer to predict one of the ten GTZAN genres.

The GraphSAGE model was trained using cross-entropy loss and the Adam optimizer. A batch size of 32 and a learning rate of 0.001 were used for 30 training epochs. The model with the highest validation accuracy was saved and later evaluated on the test set.

A CNN-based audio classifier was also used as a baseline for comparison. Unlike GraphSAGE the CNN did not use the graph structure and operated directly on the mel-spectrogram features. For the CNN input, each recording was represented using six audio segments. Recordings with fewer than six segments were zero-padded, while recordings with more than six segments were limited to the first six segments. The CNN used only the 128-bin mel-spectrogram features and did not use the 12-bin chroma features. So, each input to the CNN consisted of six segments with 128 mel features per segment. The CNN consisted of two convolutional layers with 32 and 64 filters respectively. Batch normalization and ReLU activation were applied after each convolutional layer followed by max-pooling. Adaptive average pooling was then used to obtain a fixed-size representation before the features were passed through fully connected layers. Dropout was applied to reduce overfitting and the final layer produced predictions for the ten GTZAN genres.

The CNN was trained using cross-entropy loss and the Adam optimizer with a learning rate of 0.001, a batch size of 32 and 30 training epochs. The model with the highest validation accuracy was selected as the best model and evaluated on the test set. GraphSAGE uses 140-D mel-spectrogram and chroma features and the CNN uses 128-D mel-spectrogram features only so the comparison is not strictly feature-matched.

H. Multimodal Graph-Text Fusion

The graph and text branches were combined to investigate whether structural audio information and semantic information from music descriptions could provide a more useful representation when used together. The graph branch used GraphSAGE to produce a 64-dimensional representation for each

music graph and the text branch used the 768-dimensional first-token representation from DistilBERT to represent the corresponding MusicCaps description. Two fusion strategies were implemented and trained separately: early concatenation and cross-attention.

In the early concatenation approach, the 64-dimensional graph representation was concatenated with the 768-dimensional DistilBERT text representation. The resulting multimodal representation was passed through fully connected layers with ReLU activation and dropout before the final classification layer.

The cross-attention approach allowed the graph representation to interact with the individual token representations produced by DistilBERT. The graph-level representation was projected to form a query and the DistilBERT token representations were projected to form keys and values. Scaled dot-product attention was then used to identify the text tokens most relevant to the graph representation. Padding tokens were masked so that they did not contribute to the attention computation.

The resulting attended text representation was concatenated with the graph representation and passed through a fully connected layer with ReLU activation and dropout before classification. The early-concatenation and cross-attention models were trained and evaluated independently using the same genre-based graph-text pairing strategy. The two fusion approaches were compared with the corresponding GNN-only and DistilBERT-only baselines in order to evaluate the contribution of multimodal fusion.

I. Contrastive Graph-Text Representation Learning

A dual-encoder architecture was used to learn a shared representation between music audio graphs and textual descriptions. The approach consists of separate graph and text encoders that transform audio graphs and MusicCaps descriptions into a common embedding space. The graph encoder processes the structural representation of the audio and the text encoder captures the semantic information contained in the MusicCaps description.

The graph and text representations were aligned through contrastive learning. For each paired graph-caption sample, the paired graph and caption were treated as a positive pair and non-matching graph-caption combinations within the same training batch are treated as negative pairs. This encourages the model to place the audio graphs and textual descriptions that are paired according to genre closer together in the shared embedding space and separate other graph-caption combinations. The alignment objective is based on the InfoNCE contrastive loss. Similarity between graph and text representations is measured using cosine similarity with a temperature parameter controlling the sharpness of the similarity distribution. The contrastive objective is optimized in both directions: graph-to-caption and caption-to-graph. This symmetric formulation encourages the learned embedding space to support retrieval from either modality.

After contrastive training, the learned shared embedding space is evaluated through cross-modal retrieval on the held-out test portion of the paired graph-caption dataset. Retrieval performance is measured using Recall@1, Recall@5 and Recall@10 for both caption-to-audio and audio-to-caption retrieval. Ten qualitative retrieval examples are examined by using textual descriptions as queries and retrieving the three most similar audio graphs.

The learned shared embedding space was also evaluated for zero-shot genre prediction. The ten genre labels were encoded using the same trained text encoder without any additional genre-specific training. Each test caption was then compared with the genre-label embeddings and the genre with the highest cosine similarity was selected as the zero-shot prediction. Accuracy, Macro Precision, Macro Recall and Macro-F1 were calculated and compared with the supervised multimodal classification results.

IV. EXPERIMENTAL SETUP

A. Dataset Splits and Experimental Protocol

The GTZAN and MusicCaps datasets were divided into training, validation and test subsets using an 80:10:10 split. For GTZAN, the split was performed at the graph/recording level by using stratified sampling based on the genre labels with the aim to maintain genre representation across the training, validation and test subsets. The graph file paths were also checked to ensure that no graph sample occurred in more than one subset.

For MusicCaps, the dataset was randomly shuffled and divided into training, validation and test subsets using the same 80:10:10 proportion. A fixed random seed of 42 was used to ensure reproducibility. The YouTube identifiers of the samples were checked across the three subsets to ensure that the same MusicCaps clip did not occur in multiple subsets.

For the unimodal experiments, the GTZAN splits were used for audio-based genre classification and the MusicCaps splits were used for both text-based music-tag prediction and text-based genre classification. The models were trained using the training subsets and the validation subsets were used for model selection and monitoring during training. Final performance was reported on the held-out test subsets.

For the multimodal experiments, GTZAN graph samples and MusicCaps captions were paired using the genre-based pairing strategy. Each GTZAN graph was associated with one previously unused MusicCaps caption from the same genre and the resulting graph-caption pair inherited the train, validation or test split of the GTZAN graph. So the multimodal splits were defined based on the GTZAN graph split rather than the original MusicCaps split. The multimodal dataset was constructed by genre-based pairing between GTZAN graphs and available MusicCaps captions so the resulting paired test subset contained 77 samples rather than the full 100-sample GTZAN test split. This protocol was used for both supervised multimodal fusion and contrastive graph-text representation learning.

The contrastive model was evaluated on the held-out test portion of the paired graph-caption dataset. Cross-modal retrieval was performed in both directions, using captions to retrieve audio graphs and audio graphs to retrieve captions. The same learned shared embedding space was additionally used for zero-shot genre prediction by comparing test-caption embeddings with embeddings of the ten genre labels. These zero-shot predictions were evaluated separately and compared with the supervised multimodal classification results.

B. Training Configuration

For text-based music-tag prediction, DistilBERT was trained for 5 epochs using AdamW with a learning rate of 2×10^{-5} and a batch size of 16. Binary cross-entropy with logits loss was used with 50 music tags as prediction targets. For text-based genre classification, the DistilBERT-based classifier was trained for 20 epochs using Adam with a learning rate of 0.0001 and a batch size of 8. Cross-entropy loss was used for the ten genre classes. For audio-based genre classification, both the CNN and GraphSAGE models were trained for 30 epochs using Adam with a learning rate of 0.001 and a batch size of 32. Cross-entropy loss was used for the ten genre classes. For early-concatenation fusion, the model was trained for 20 epochs using Adam with a learning rate of 0.0001 and a batch size of 8. Cross-entropy loss was used, and DistilBERT was fine-tuned. For cross-attention fusion, the model was trained for 20 epochs using AdamW with a learning rate of 0.001 and a batch size of 4. Cross-entropy loss was used, and DistilBERT was kept frozen. For contrastive graph-text learning, the dual encoder was trained for 20 epochs using AdamW with a learning rate of 0.001 and a batch size of 8. Graph and text representations were projected into a shared 256-dimensional normalized embedding space. Symmetric InfoNCE loss with a temperature of 0.07 was used and DistilBERT was kept frozen.

C. Evaluation Metrics

The performance of the models was evaluated on the test set using several classification and retrieval metrics. These metrics were used to compare the baseline models, fusion models and ablation models where applicable.

For text-based music-tag prediction, Macro-F1, Micro-F1, Macro Precision, Macro Recall and test loss were used to evaluate multi-label classification performance. For genre classification experiments, accuracy, precision, recall, Macro-F1, and where applicable, Micro-F1, AUC-PR and test loss were used. Accuracy measures the percentage of correctly classified samples and precision and recall measure the correctness and completeness of the genre predictions. Macro-F1 gives equal importance to all genres by averaging their individual F1-scores and Micro-F1 measures the overall performance across all samples. AUC-PR evaluates the model's performance based on the precision-recall relationship and test loss measures the error between the predicted probability distribution and the actual class labels.

For cross-modal retrieval, Recall@1, Recall@5 and Recall@10 were used. These metrics measure whether the correct audio or caption appears within the top 1, 5 or 10 retrieved results. Retrieval performance was measured in both caption-to-audio and audio-to-caption directions. For the zero-shot genre experiment, accuracy, Macro Precision, Macro Recall and Macro-F1 were used. These metrics evaluate whether the dual encoder can identify the correct genre without being directly trained on genre labels.

D. Hardware and Software

The project was developed and evaluated on a Windows-based computer using Python 3.11.9. PyTorch was used for model development and PyTorch Geometric was used for graph processing and GNN models. Hugging Face Transformers was used for DistilBERT-based text processing. Librosa was used for audio processing and NumPy, Pandas and Scikit-learn were used for data processing and evaluation. The experiments were run using the available CPU of the system.

V. RESULTS AND ANALYSIS

A. Text-Based Music Tag Prediction

A DistilBERT-based multi-label classifier was evaluated for understanding textual music context without using graph information. The model was trained to predict music-related tags from captions and evaluated on the test set using Macro-F1, Micro-F1, Macro Precision, Macro Recall and test loss.

TABLE I
TEST PERFORMANCE OF THE DISTILBERT-BASED MULTI-LABEL CLASSIFIER.

Metric	Test Score
Macro-F1	0.5153
Micro-F1	0.6930
Macro Precision	0.6690
Macro Recall	0.4722
Test Loss	0.0955

In Table I, the model achieved a Micro-F1 of 0.6930 which indicates that it was able to identify many of the relevant tags across the test samples. The lower Macro-F1 of 0.5153 is influenced likely by the imbalance between frequently occurring and less frequent tags. For example, tags such as low quality and instrumental occur much more frequently than less common tags such as e-guitar. The Macro Precision of 0.6690 compared with a Macro Recall of 0.4722 indicates that the model’s predicted tags were reliable generally even though it missed a number of relevant tags.

Fig. 1 shows that the validation Macro-F1 and Micro-F1 scores increased consistently across the five training epochs which indicates that the model’s tag prediction performance improved during fine-tuning. The Micro-F1 remained higher than Macro-F1 which shows stronger overall performance on frequently occurring tags than across all tags equally.

In Table II, the examples show that the model was able to identify many meaningful semantic tags from the music

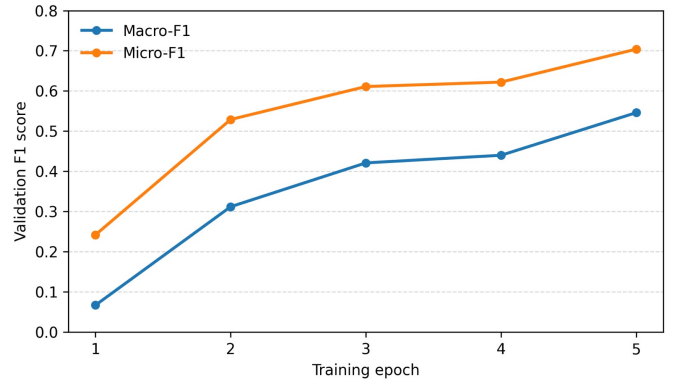


Fig. 1. Validation Macro-F1 and Micro-F1 across the five DistilBERT fine-tuning epochs.

TABLE II
EXAMPLE MUSIC-TAG PREDICTIONS FROM THE DISTILBERT BASELINE.

Example	Actual Tags	Predicted Tags
1	instrumental	None
2	medium tempo, amateur recording, acoustic guitar	amateur recordings
3	low quality, noisy, mono	low quality, noisy, mono
4	low quality, groovy, punchy kick, punchy snare, shimmering hi hats	low quality, punchy kick, punchy snare, shimmering hi hats
5	low quality, punchy kick, groovy bass, pop, easygoing	low quality, punchy kick, punchy snare, easygoing

captions. Example 3 predicted correctly all three ground-truth tags and Examples 2, 4, 5 contained many correct predictions but also missed some tags. In Example 1, no tag was predicted because the model’s confidence for the available labels remained below the classification threshold. The examples show that the DistilBERT baseline can capture useful semantic information from textual music descriptions and this provides a text-based baseline for comparison with the graph and multimodal models later on.

B. Graph-Based Music Genre Classification

A GraphSAGE model was evaluated for genre classification using the constructed audio segment graphs from the GTZAN dataset. Each graph represents a track using audio-based node features which allows the model to learn from relationships between different segments. A CNN baseline using Mel-spectrogram features was also evaluated for comparison. The experiment evaluated whether the graph representation could capture useful genre information compared with a conventional CNN-based representation.

In Table III, the GraphSAGE model achieved a test accuracy of 66.0% and a Macro-F1 of 65.9% which shows that the graph representation was able to capture meaningful genre information. There are ten genres so random classification would give an accuracy of approximately 10% and so the GraphSAGE performance is already above this baseline. The Macro Precision and Macro Recall were 67.1% and 66.0%

TABLE III
GENRE CLASSIFICATION PERFORMANCE OF GRAPH SAGE AND CNN
BASELINE.

Metric	GraphSAGE	CNN
Test Accuracy	66.0%	73.0%
Macro Precision	67.1%	78.2%
Macro Recall	66.0%	73.0%
Macro-F1	65.9%	71.6%

respectively which shows that the model had similar overall precision and recall when averaged across the genres.

The per-genre results show that classical, jazz and metal were the strongest categories for GraphSAGE and they had F1-scores of 0.89, 0.82 and 0.76 respectively. But rock, reggae and hip-hop were more difficult as they had F1-scores of 0.53, 0.56 and 0.57. This suggests that genres with more distinguishable acoustic characteristics were easier for the graph model to classify and genres with more similar acoustic characteristics were more difficult to distinguish.

The CNN baseline achieved higher performance as it had 73.0% accuracy and 71.6% Macro-F1. It also achieved higher Macro Precision and Macro Recall than GraphSAGE. So the CNN achieved higher performance than GraphSAGE under the evaluated configurations. However the difference in input features should be considered when interpreting this comparison. The GraphSAGE results still show that the constructed graph representation contains useful information for genre classification and provides a basis for the graph branch that is used in the multimodal experiments.

C. GNN-BERT Multimodal Fusion

This experiment evaluates whether combining structural information from audio graphs with semantic information from MusicCaps captions can improve genre classification and multimodal representation learning. The graph branch uses a GraphSAGE model and the text branch uses a DistilBERT-based classifier. The experiment evaluates multimodal genre-level alignment rather than exact audio-caption correspondence.

TABLE IV
ABLATION RESULTS.

Model	Accuracy	Macro-F1	Macro AUC-PR
GNN-only	72.73%	62.18%	66.62%
DistilBERT-only	76.62%	67.97%	78.87%
Early Concat	80.52%	73.43%	77.71%
Cross-Attention	80.52%	73.79%	91.88%

Note that the GNN-only performance here reflects the specific paired multimodal test subset which differs from the full unimodal GTZAN evaluation. The fusion models also used different training configurations so the performance difference cannot be attributed solely to the fusion mechanism.

In Table IV, both fusion methods outperformed the individual models in accuracy and Macro-F1 within the paired multimodal test subset. The GNN-only model achieved 72.73% accuracy and 62.18% Macro-F1 and the DistilBERT-only model

achieved 76.62% accuracy and 67.97% Macro-F1. Combining the two modalities by using early concatenation increased the performance to 80.52% accuracy and 73.43% Macro-F1. The cross-attention model achieved the same accuracy but a little higher Macro-F1 of 73.79%.

A larger difference can be seen in Macro AUC-PR. Early concatenation achieved 77.71% and cross-attention achieved 91.88%. This shows that even though both fusion methods produced the same accuracy yet cross-attention achieved higher Macro AUC-PR which indicates better precision-recall discrimination across the genre classes. This suggests that cross-attention may provide a more effective multimodal representation than direct concatenation even though the difference cannot be attributed solely to the fusion mechanism.

The final cross-attention model achieved 80.52% test accuracy, 73.79% Macro-F1, 80.52% Micro-F1 and 91.88% Macro AUC-PR. The Micro AUC-PR was 89.61% and had a test loss of 0.8451. The model performed quite well on country and metal and both had an F1-score of 0.95. Classical and hip-hop achieved 0.82 and jazz and rock achieved 0.78 and pop achieved 0.74. Blues and reggae had high recall but lower precision and they had fewer test samples in these classes. The test set contained 77 samples and no disco samples. So the zero support for disco should not be interpreted as poor disco performance. The uneven class distribution should also be considered when interpreting the macro-level metrics.

The 192-dimensional fused representations were visualized by using t-SNE to examine their grouping patterns by genre and mood.

In Fig. 2, the genre visualization shows relatively distinct clustering between several genres. Groups such as classical, jazz, metal, country, hip-hop and pop form different regions of the embedding space. This suggests that the fused representation captures useful genre-related information. In contrast, the mood visualization shows greater overlap between categories. The samples from the same mood were distributed across multiple regions of the embedding space. So the fused representation appears to organize the samples more strongly according to genre than mood. The mood labels were derived from the semantic annotations so these patterns should be interpreted more cautiously.

Three qualitative examples were also examined to illustrate the interaction between the audio graph and textual information. The fusion model predicted the genre correctly in all three cases.

Metal: The model identified the metal genre correctly. The caption contains terms related to metal, guitars, drumming, cymbals and an energetic sound which provides semantic information that is consistent with the predicted genre.

Classical: The model identified the classical genre correctly. The caption describes strings, flute and a live classical performance and the attention visualization gives strong weight to the genre-related caption information.

Jazz: The model identified the jazz genre correctly. The caption includes jazz, electric guitar, bass guitar, acoustic

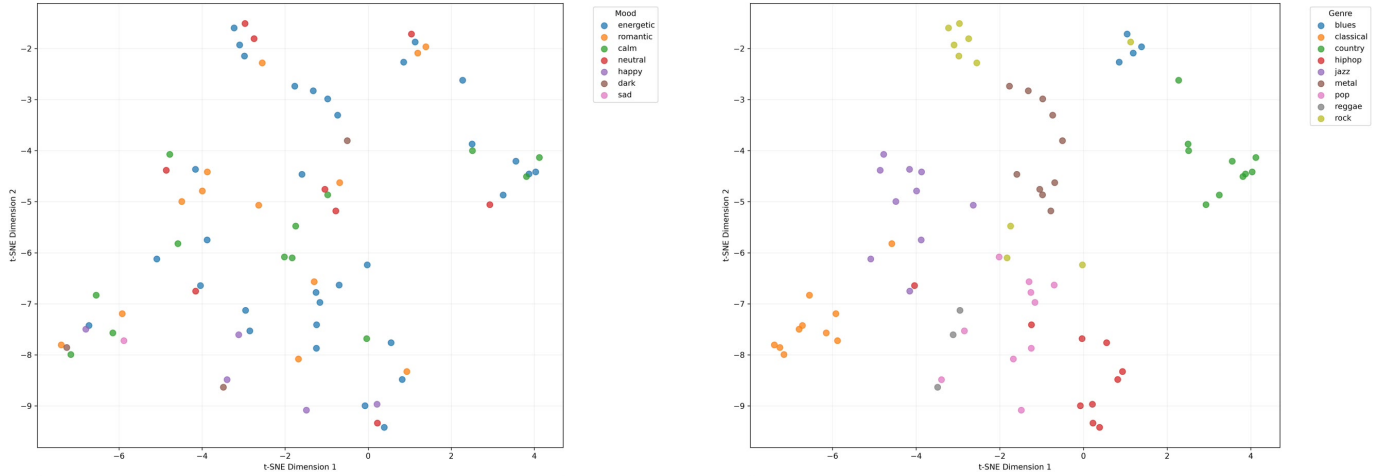


Fig. 2. t-SNE visualization of the fused GNN-BERT representations on the test set, colored by mood and genre.

drums and a chill atmosphere which provides textual context that is consistent with the predicted genre.

These examples show how the fusion model combines graph-based audio information with relevant textual context for genre prediction.

D. Cross-Modal MusicCaps Alignment

A contrastive cross-modal retrieval task was implemented by using a GNN-BERT dual encoder which is trained with symmetric InfoNCE loss. Retrieval performance was evaluated by using Recall@1, Recall@5 and Recall@10 in both caption-to-audio and audio-to-caption directions.

The model achieved a best validation contrastive loss of 1.6369 and a test contrastive loss of 1.6442.

TABLE V
CROSS-MODAL RETRIEVAL PERFORMANCE OF THE DUAL ENCODER

Direction	R@1	R@5	R@10
Caption → Audio	6.49%	28.57%	42.86%
Audio → Caption	7.79%	24.68%	40.26%

In Table V, the dual encoder achieved Recall@1 values of 6.49% and 7.79% for caption-to-audio and audio-to-caption retrieval respectively. Performance increased at higher retrieval thresholds as it reached 42.86% and 40.26% at Recall@10. With 77 test samples the corresponding random-retrieval baselines are 1.30%, 6.49% and 12.99% for R@1, R@5 and R@10 approximately. So the model performs above random retrieval in both directions especially at R@10. Caption-to-audio retrieval performs better at R@5 and R@10 and audio-to-caption retrieval has a little higher R@1. The results suggest that the contrastive model learned useful cross-modal relationships especially at larger retrieval thresholds but lower R@1 values show that identifying the exact matching item is still challenging.

The ten qualitative retrieval examples showed that the shared embedding space captured several meaningful cross-modal similarities. Retrieved samples shared characteristics

frequently such as instrumentation, rhythm, tempo, mood or genre-related information with the query caption but genre-level agreement was not always consistent. For example, a contemporary rap/hip-hop query retrieves two hip-hop samples within the top two positions and the classical query retrieves three classical samples. But some queries retrieved samples from related but different genres such as the jazz and blues queries. These examples suggest that the learned representation captures broader semantic similarities even when exact retrieval is difficult. The complete set of ten qualitative retrieval examples including the top-three retrieved clips and similarity scores is provided in the project results.

The learned shared embedding space was also evaluated for zero-shot genre prediction by comparing caption embeddings with embeddings of the ten genre labels. Zero-shot tag prediction was also evaluated but performance was limited by the large and fine-grained tag vocabulary. So zero-shot genre prediction is reported as the primary zero-shot classification analysis because it provides a more direct comparison with the ten-class supervised model.

TABLE VI
COMPARISON OF SUPERVISED FUSION AND ZERO-SHOT GENRE PREDICTION

Evaluation	Accuracy	Macro Precision	Macro Recall	Macro-F1
Fusion Model	80.52%	75.44%	75.00%	73.79%
Zero-shot Genre	18.18%	23.44%	22.50%	14.90%

In Table VI, the supervised fusion model outperformed the zero-shot genre approach across all evaluated metrics. The zero-shot genre prediction performed above the approximately 10% random-guess accuracy for ten genres which shows that the shared embedding space captured some genre-related information without using genre labels directly in the contrastive loss. Its lower performance is likely related to genre-based graph-caption pairing rather than exact recording-level correspondence as well as the difficulty of comparing rich caption embeddings with embeddings of individual genre

labels. This large performance gap highlights the difference between learning genre-specific decision boundaries through supervised classification and inferring genre similarity directly from a shared embedding space.

The zero-shot approach also achieved 45.45% Top-3 accuracy and 58.44% Top-5 accuracy which indicates that the correct genre was often present among its higher-ranked predictions.

VI. DISCUSSIONS AND LIMITATIONS

Overall the results show that graph-based and text-based representations capture complementary information about music and their combination provides the strongest supervised performance. The GraphSAGE model achieved reasonable genre classification even though the CNN baseline performed better which suggests that the current graph representation does not capture acoustic information as effectively as the spectrogram-based representation. DistilBERT also captured useful semantic information from music captions. Both early concatenation and cross-attention improved performance when graph and text representations were combined as cross-attention provided the strongest precision-recall performance. The t-SNE visualizations suggest that the fused representation organizes the data more clearly by genre than by mood although the mood labels were derived heuristically from semantic annotations and so should be interpreted cautiously. The contrastive dual-encoder also achieved retrieval performance above random which shows that it learned some meaningful cross-modal relationships but the lower retrieval scores show that exact graph-caption alignment is challenging. The main limitations are the genre-based pairing of GTZAN graphs and MusicCaps captions rather than exact recording-level correspondence, the uneven test-class distribution including the absence of Disco samples in the test set and some evidence of overfitting during fusion training. Zero-shot tag prediction was also limited by the large and fine-grained candidate vocabulary and lexical variation in the tags so zero-shot genre prediction provides the more direct zero-shot comparison. The multimodal evaluation was limited by the smaller 77-sample paired test set. There were also some differences between the model settings such as using 140-D features for GraphSAGE and 128-D features for the CNN and using different fine-tuning configurations for the fusion models. These differences should be considered when interpreting the results.

VII. CONCLUSION

This study demonstrated the potential of combining graph-based and language-based representations for music context understanding. The proposed GNN-BERT framework showed that structural information from music graphs and semantic information from textual descriptions can complement each other as multimodal fusion achieved stronger supervised performance than the individual representations. The contrastive learning approach also showed the ability to learn meaningful relationships between audio graphs and textual descriptions through a shared embedding space. There were limitations

in the current graph-caption pairing and dataset size but the results support the potential of combining structural and semantic representations for multimodal music understanding. Future work could focus on instance-level audio-text pairing, larger and more balanced datasets and richer graph representations in order to improve multimodal alignment and retrieval performance.

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